

RESEARCH METHODS

DAY 4

Welcome to Day 4!

Let's start with a fun Quiz!

And refresh our memories on source credibility.

Question 1

True or False: A personal blog is a reliable source for scientific facts.

False!

Question 2

Which of the following is usually the MOST reliable for accurate scientific information?

- A. Wikipedia
- B. TikTok video
- C. Peer-reviewed science journal
- D. Reddit post

Answer: C

Question 3

What should you check to see if a website is trustworthy?

- A. Its color and design
- B. If it has ads
- C. Who wrote it and when it was last updated
- D. How many emojis it uses

Answer: C

Question 4

Which source is the least likely to be biased?

- A. A product review written by the company that made the product
- B. A university study published in a journal
- C. A celebrity's opinion on a diet trend
- D. A tweet from a random user

Answer: B

Question 5

True or False: An article that includes sources and references is more trustworthy than one that doesn't.

True!

Question 6

True or False: All information on the internet is true if it has lots of likes or views.

False!

Question 7

True or False: It's okay to use Wikipedia as a starting point, but you should double-check the facts with more reliable sources.

False!

True!!!

Question 8

Which of these sources would most likely be considered a primary source?

- A. A YouTube video about World War II
- B. A soldier's diary from World War II
- C. A news article written last week about World War II
- D. A textbook chapter on World War II

Answer: B

Question 9

Which of these sources is usually written by experts and reviewed before being published?

- A. Peer-reviewed journal
- B. Instagram story
- C. Student opinion piece
- D. Online ad

Answer: A

Question 10

Which of these shows bias in a source?

- A. The author includes only one side of an argument
- B. The source shows both pros and cons
- C. The author lists sources at the end
- D. The source has a publication date

Answer: A

Bonus Question

True or False: A website with lots of ads and pop-ups might not be a good place to find trustworthy information.

True!

Reliable vs. Unreliable Sources

Reliable Sources

- Backed by verifiable evidence
- Created by experts
- Undergo review processes
- Transparent about methods
- Cite their sources

Unreliable Sources

- Lack transparency
- Anonymous or unclear authorship
- Sensationalist language
- One-sided arguments
- Missing citations

NEW!

Examples of Reliable Sources



CDC

Centers for Disease Control and Prevention



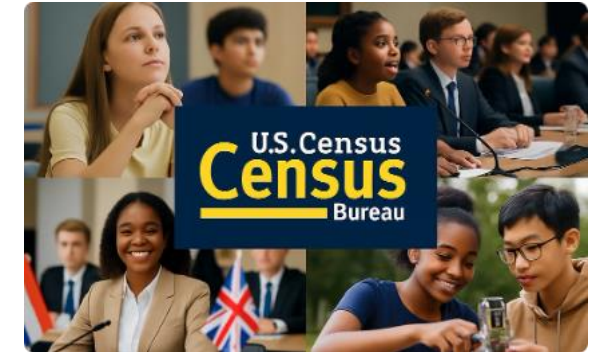
NASA

National Aeronautics and Space Administration



NEJM

Academic Medical Journal



US Census Bureau

Federal Statistical Agency



WHO

World Health Organization



National Geographic

Science and Education Publication



Smithsonian

Research and Museum Institution

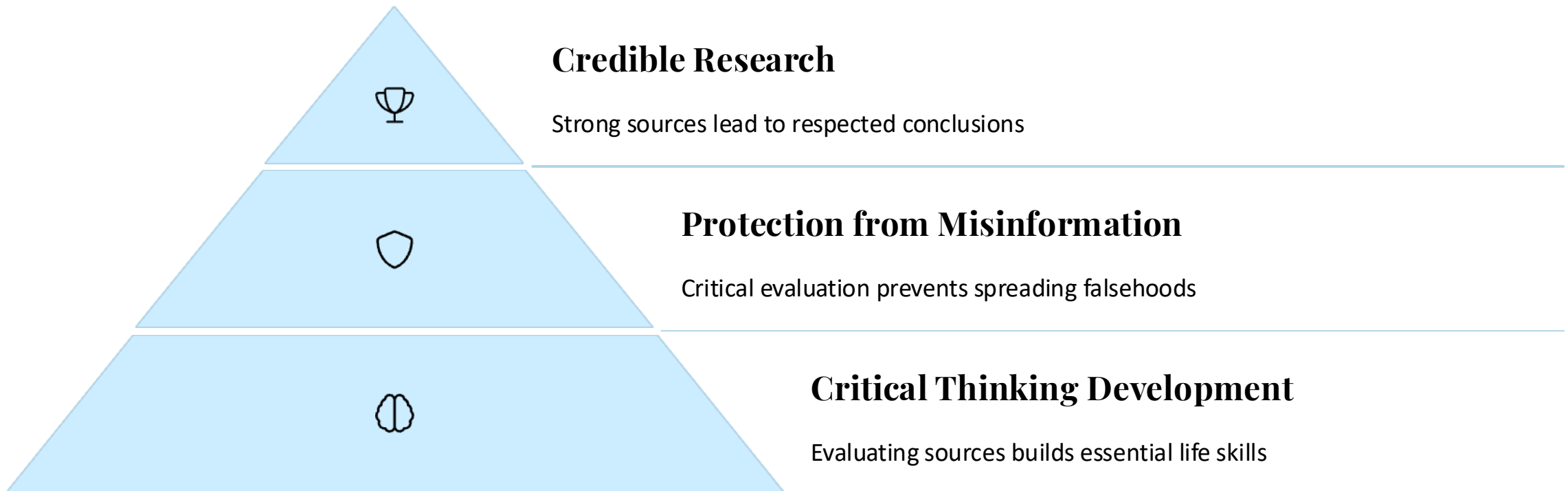


APA

American Psychological Association

These institutions are known for rigorous research methods, expert review processes, and commitment to accuracy in their published information.

Why Source Evaluation Matters



Red Flags of Misinformation



Sensational Language

"You won't believe this!" or excessive punctuation for shock value



Missing Attribution

No clear author, source, or supporting evidence



One-Sided Presentation

No acknowledgment of alternative perspectives or debates



Extraordinary Claims

Bold statements without proportional evidence

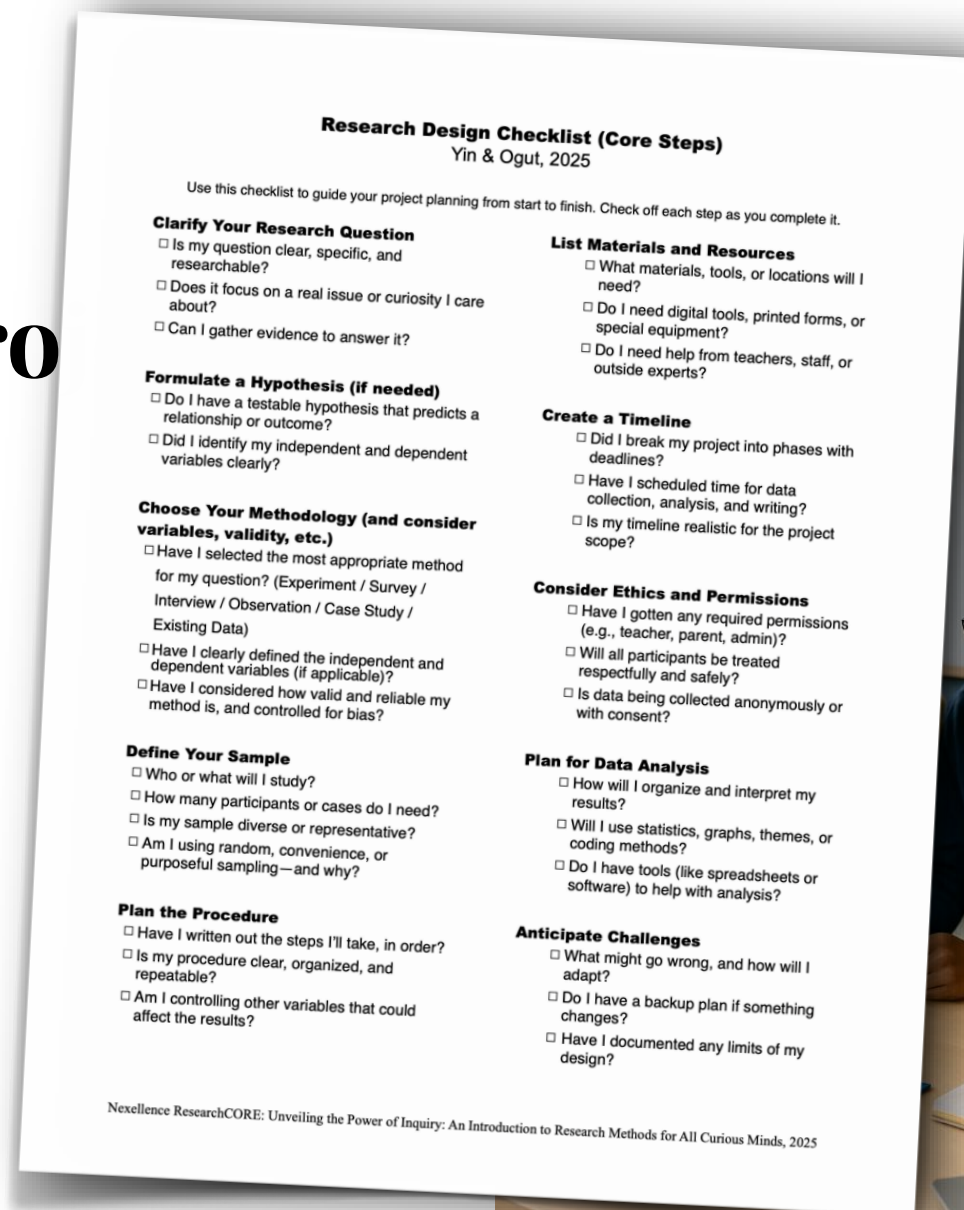


Designing a Research Project



Designing a Research Pro

To make the steps of designing easier, you can review & follow our checklist.



Designing a Research Pro

Look where we are in the process...



Research Design Checklist (Core Steps)
Yin & Ogut, 2025

Use this checklist to guide your project planning from start to finish. Check off each step as you complete it.

Clarify Your Research Question ☐ Is my question clear, specific, and researchable? ☐ Does it focus on a real issue or curiosity I care about? ☐ Can I gather evidence to answer it? Formulate a Hypothesis (if needed) ☐ Do I have a testable hypothesis that predicts a relationship or outcome? ☐ Did I identify my independent and dependent variables clearly? Choose Your Methodology (and consider variables, validity, etc.) ☐ Have I selected the most appropriate method for my question? (Experiment / Survey / Interview / Observation / Case Study / Existing Data) ☐ Have I clearly defined the independent and dependent variables (if applicable)? ☐ Have I considered how valid and reliable my method is, and controlled for bias? Define Your Sample ☐ Who or what will I study? ☐ How many participants or cases do I need? ☐ Is my sample diverse or representative? ☐ Am I using random, convenience, or purposeful sampling—and why? Plan the Procedure ☐ Have I written out the steps I'll take, in order? ☐ Is my procedure clear, organized, and repeatable? ☐ Am I controlling other variables that could affect the results?	List Materials and Resources ☐ What materials, tools, or locations will I need? ☐ Do I need digital tools, printed forms, or special equipment? ☐ Do I need help from teachers, staff, or outside experts? Create a Timeline ☐ Did I break my project into phases with deadlines? ☐ Have I scheduled time for data collection, analysis, and writing? ☐ Is my timeline realistic for the project scope? Consider Ethics and Permissions ☐ Have I gotten any required permissions (e.g., teacher, parent, admin)? ☐ Will all participants be treated respectfully and safely? ☐ Is data being collected anonymously or with consent? Plan for Data Analysis ☐ How will I organize and interpret my results? ☐ Will I use statistics, graphs, themes, or coding methods? ☐ Do I have tools (like spreadsheets or software) to help with analysis? Anticipate Challenges ☐ What might go wrong, and how will I adapt? ☐ Do I have a backup plan if something changes? ☐ Have I documented any limits of my design?
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PROJECT DESIGN



Failing to Plan is Planning to Fail



Research Requires Direction

Like baking without a recipe or traveling without a map, research without planning leads to confusion.



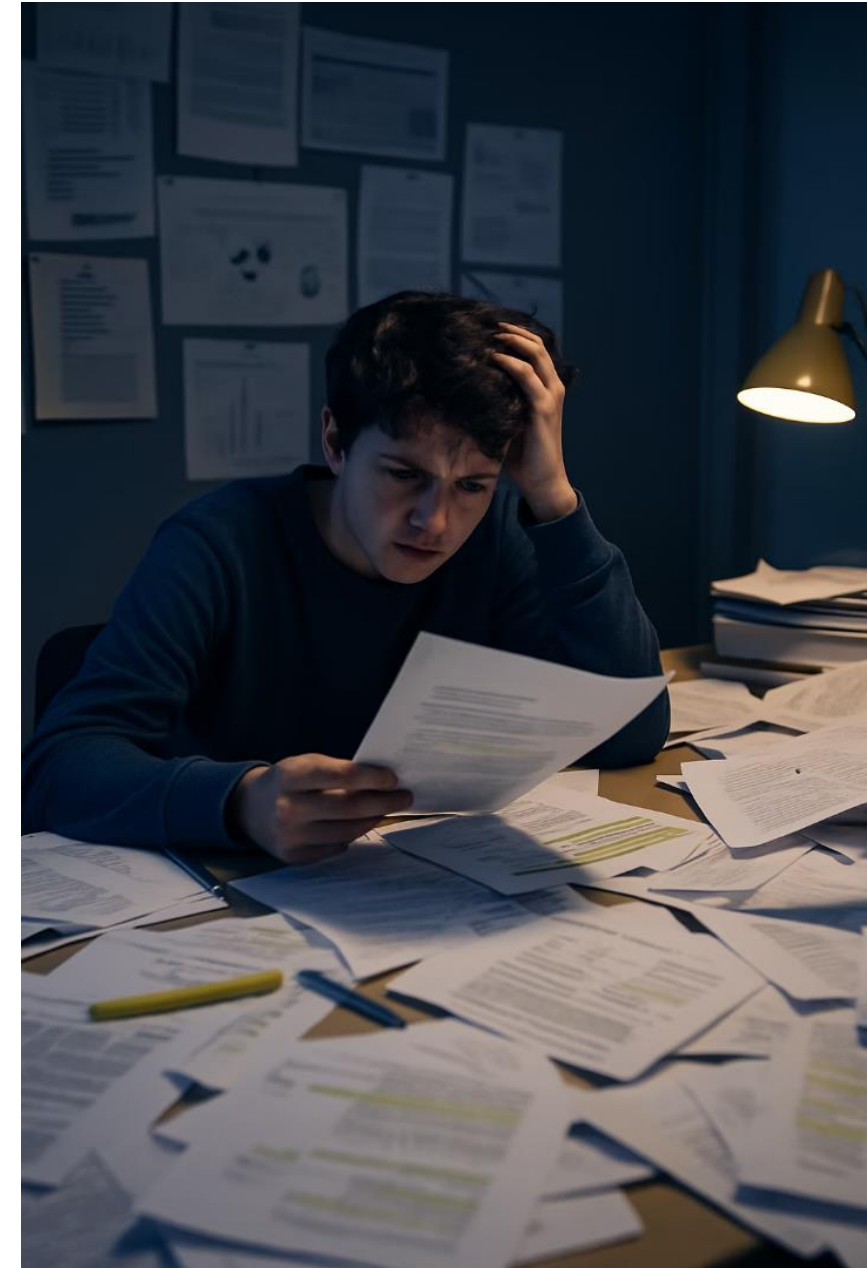
Create a Blueprint

A well-designed project saves time and increases the chances of valid, meaningful findings.



Develop a Game Plan

We'll cover methodologies, tools, variables, and ethical considerations for your research journey.



Choosing a Research Method

The first step in designing your project is selecting the right method to answer your question.

Methodology includes...

- Your overall approach / strategy:
Qualitative. Quantitative. Mixed.
- Your data collection tools / techniques
 - Experiment
 - Surveys
 - Interview
 - Focus Groups
 - Observation
 - Case Study
 - Mixed Methods

(We talked about the first four methods of collecting your own data yesterday)

RESEARCH METHODS



QUANTITATIVE



QUALITATIVE

Observation: Watching Real Behavior



Natural Settings

Observe behavior as it naturally occurs without intervention.



Structured or Unstructured

Use checklists for specific behaviors or take open-ended notes on everything you see.



Time Sampling

Observe at different times to capture variation in behavior patterns.



Minimize Observer Effect

Be discreet so your presence doesn't change the behavior you're studying.

Can be utilized in quantitative or qualitative studies

Case Studies: Deep Dives

Definition

Intensive study of a single case or small number of cases to gain in-depth understanding.

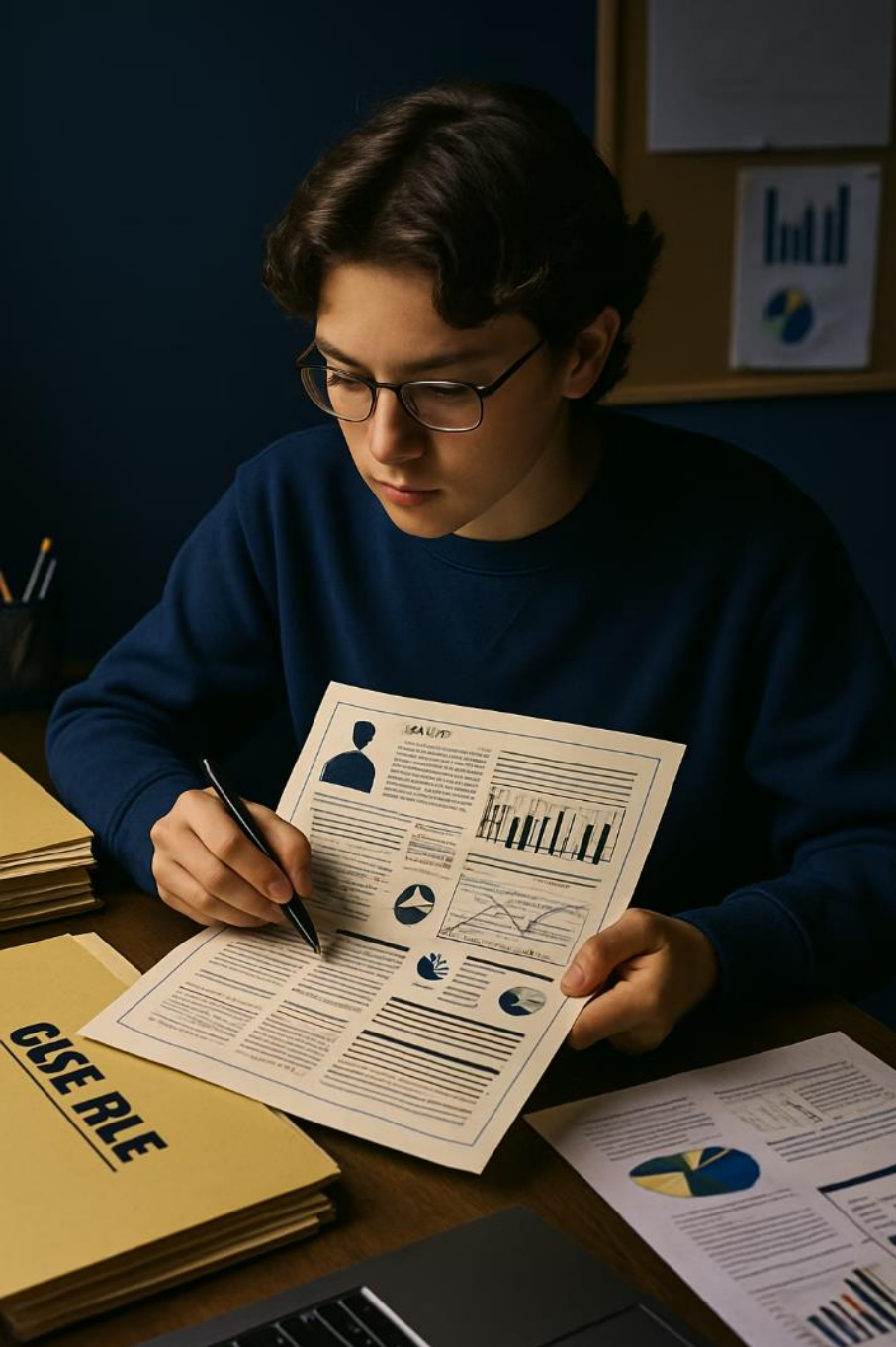
When to Use

For unique situations, rare phenomena, or when you need comprehensive understanding of one example.

Limitations

Findings may not generalize to other cases or populations.

Mostly qualitative but can include quantitative elements



Using Existing Data: Secondary Research



Efficient Access

Use data already collected by others instead of gathering new information.



Historical Perspective

Analyze trends over time periods you couldn't study directly.



Multiple Sources

Government reports, research studies, archives, statistics, and historical documents.



Source Verification

Always evaluate the credibility and quality of secondary data sources.

Mixed Methods: Combining Approaches & Tools

Benefits

- Provides fuller picture of the topic
- Combines breadth and depth
- Numbers + stories = comprehensive understanding
- Compensates for weaknesses in single methods

Examples

- Survey + interviews
- Experiment + questionnaire
- Observation + focus groups
- Case study + statistical analysis



Mixed methods require careful planning but provide rich, multidimensional insights.

At camp, we are to conduct secondary research.
What could you at a parallel universe or simply another time?

Let's team brainstorm!

Here is the question to answer: If we were to answer your research question from scratch (no existing data), how could you go about collecting the data?

Shall we play a game?

Method Match

Choosing the Right Method



Consider your research question

What type of answer do you need? Numbers, descriptions, or both?



Assess your resources

How much time, access, and materials do you have available?



Think about participants

Who will you study and how many can you realistically include?



Weigh strengths and limitations

Each method has trade-offs. Choose what best fits your specific needs.

Correlation vs. Causation

Correlation

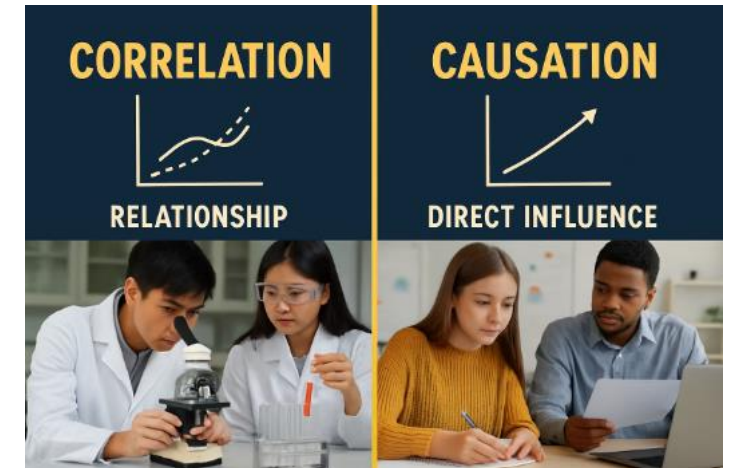
Two variables change together or are related.

- Shows a pattern or relationship
- Doesn't prove one causes the other
- Found in surveys, observations, etc.

Causation

One variable directly causes changes in another.

- Shows cause and effect
- Requires controlled experiments
- Harder to establish definitively



*Remember:
"Correlation does not equal
causation" is a fundamental
principle in research.*

The Shoe Size-Reading Ability Example

Kids with bigger feet tend to read better.

Variable 1: Shoe Size

Variable 2: Reading Ability

True Cause: Age

Age affects both shoe size and reading ability



Planning the Procedure

Once you've chosen a method,
you need a detailed step-by-step plan
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Planning the Procedure

Once you've chosen a method, you need a detailed step-by-step plan for your research.

Write Clear Instructions

Create a detailed "recipe" for your research that someone else could follow.

Consider All Variables

Identify what you'll change, what you'll measure, and what you'll keep constant.

Plan Data Collection

Decide exactly what data you'll gather and how you'll record it.

Anticipate Challenges

Think about what might go wrong and how you'll handle it.

Experiment Procedure Planning



Define Your Groups

Decide who or what will be in experimental and control groups.
Use random assignment when possible.



Define the Treatment

Clearly describe what you'll do with the experimental group. Be specific enough for replication.



Define Control Activities

Decide what the control group will do during the same time period.



Plan Data Collection

Determine exactly what measurements you'll take, when, and how.



Survey Procedure Planning

Distribution Method

Decide how to give out your survey: paper, online, in-person. Each has pros and cons for reach and response rate.

Target Respondents

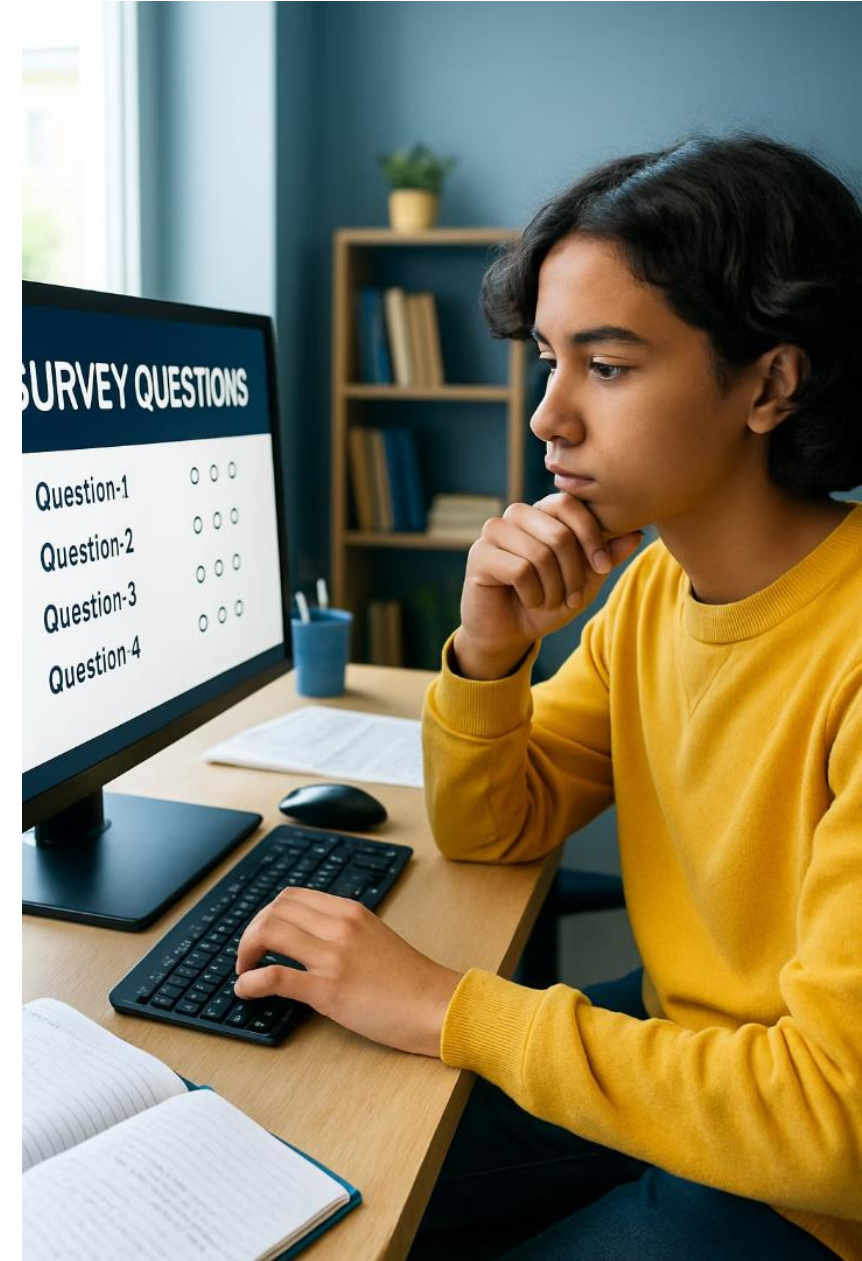
Define who should take your survey. Aim for a representative sample, not just friends.

Question Design

Write clear, unbiased questions. Test them on a few people first to check for confusion.

Response Collection

Plan how to gather and analyze answers. Set a timeline with reminders for participants.



Observation Procedure Planning



Choose Locations and Times

Create a clear schedule for when and where you'll observe. Include multiple times to identify patterns.



Define Observation Focus

Decide exactly what behaviors or phenomena you'll look for to stay organized.

3

Plan to Minimize Influence

Be discreet so your presence doesn't change the behavior you're studying.



Create Recording Method

Decide how you'll document observations: tally sheet, notes, sketches, or digital tools.



Using Existing Data: Procedure Planning



Identify Sources

Make a list of all data sources you plan to use, like databases, archives, or publications.



Verify Credibility

Check that sources are trustworthy, like government agencies, universities, or reputable organizations.



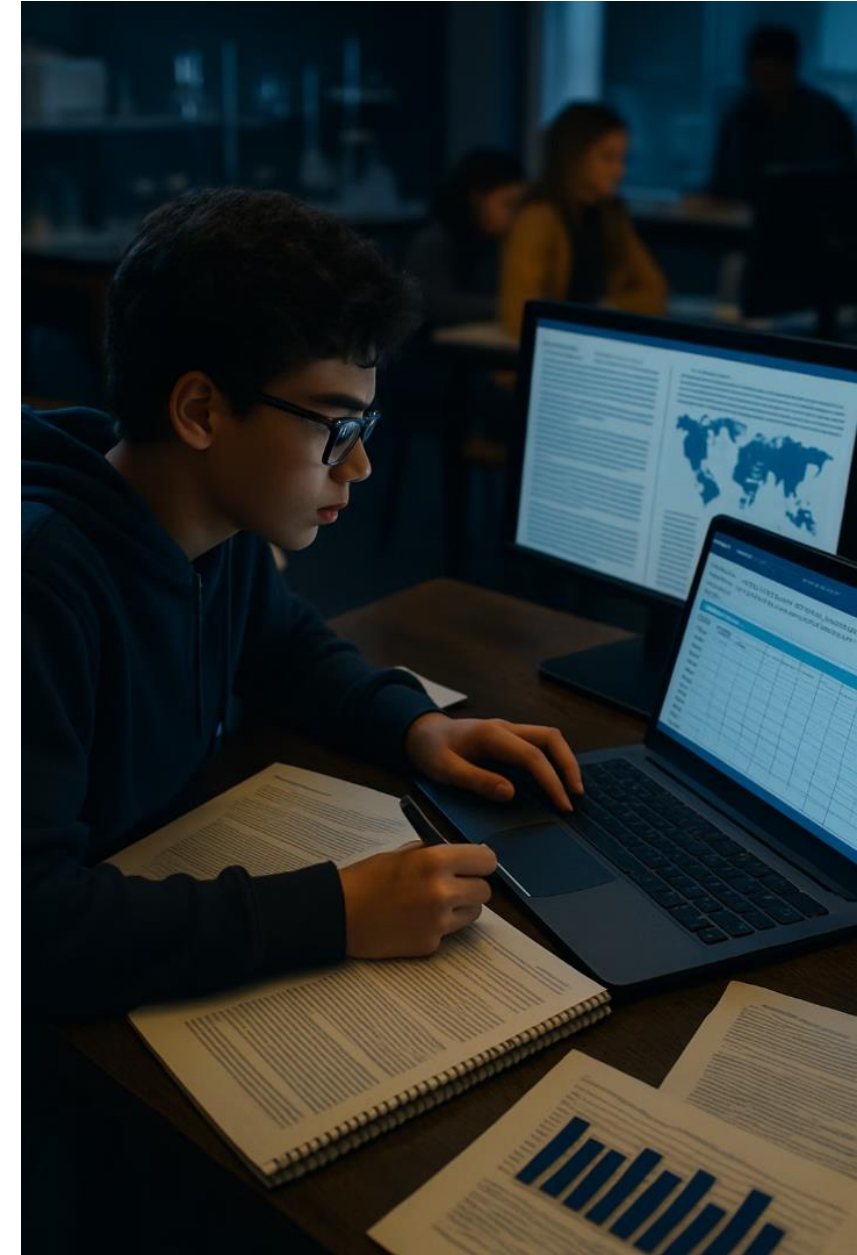
Plan Data Extraction

Decide what specific information you need from each source and how you'll organize it.



Organize Information

Create a system to track where each piece of data came from and how it relates to your question.



Tomorrow we continue to look into planning

Specifically reflecting on validity & reliability, sampling, and materials/resources.



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- ☐ Is my question clear, specific, and researchable?
- ☐ Does it focus on a real issue or curiosity I care about?
- ☐ Can I gather evidence to answer it?

List Materials and Resources

- ☐ What materials, tools, or locations will I need?
- ☐ Do I need digital tools, printed forms, or special equipment?
- ☐ Do I need help from teachers, staff, or outside experts?

Formulate a Hypothesis (if needed)

- ☐ Do I have a testable hypothesis that predicts a relationship or outcome?
- ☐ Did I identify my independent and dependent variables clearly?

Create a Timeline

- ☐ Did I break my project into phases with deadlines?
- ☐ Have I scheduled time for data collection, analysis, and writing?
- ☐ Is my timeline realistic for the project scope?

Choose Your Methodology (and consider variables, validity, etc.)

- ☐ Have I selected the most appropriate method for my question? (Experiment / Survey / Interview / Observation / Case Study / Existing Data)
- ☐ Have I clearly defined the independent and dependent variables (if applicable)?
- ☐ Have I considered how valid and reliable my method is, and controlled for bias?

Consider Ethics and Permissions

- ☐ Have I gotten any required permissions (e.g., teacher, parent, admin)?
- ☐ Will all participants be treated respectfully and safely?
- ☐ Is data being collected anonymously or with consent?

Define Your Sample

- ☐ Who or what will I study?
- ☐ How many participants or cases do I need?
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- ☐ Am I using random, convenience, or purposeful sampling—and why?

Plan for Data Analysis

- ☐ How will I organize and interpret my results?
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Plan the Procedure

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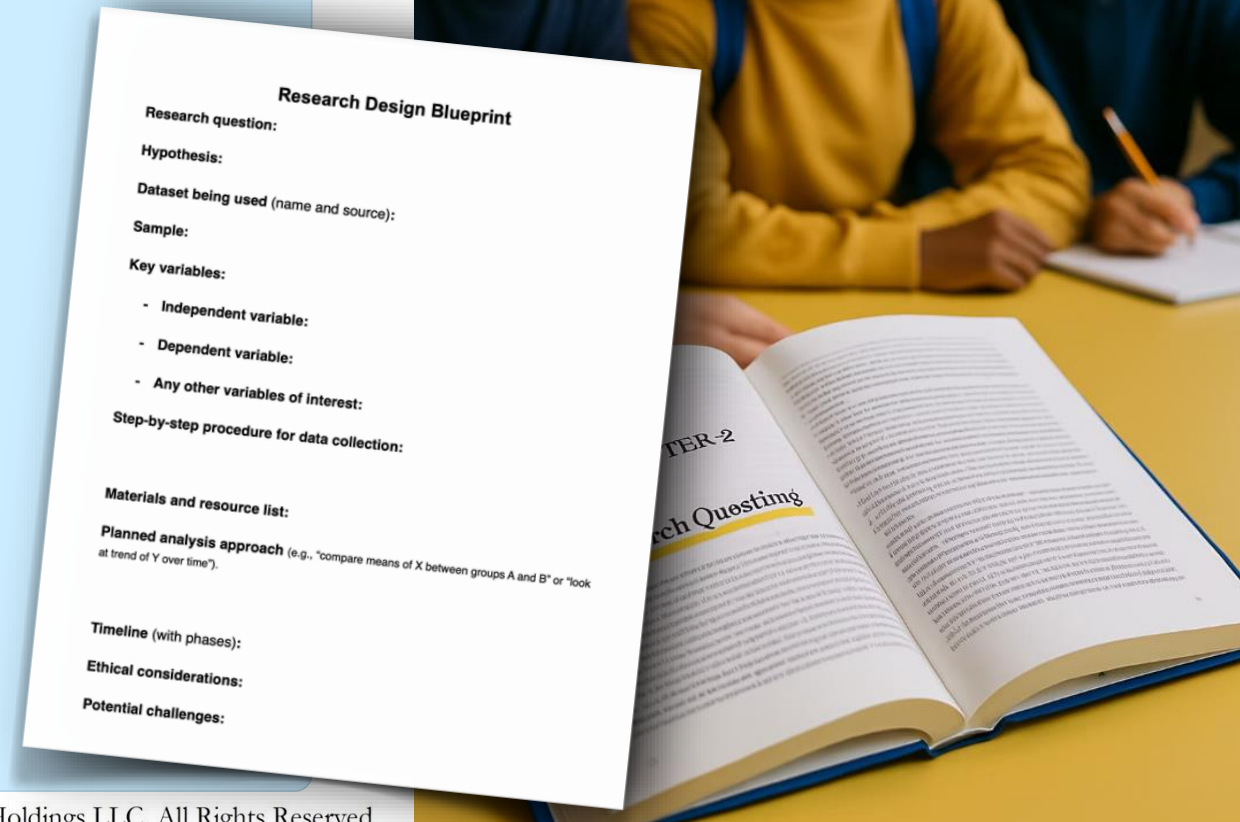
Anticipate Challenges

- ☐ What might go wrong, and how will I adapt?
- ☐ Do I have a backup plan if something changes?
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Homework - Teamwork

- Review: "Research Design Checklist" with your team members.
- Start working on Handout 4: Research Design Blueprint. Finish first half of the items.
- Continue with your lit review. Review 1-2 more articles and keep track of your resources through "Source Tracker". Identify 2-3 of the most important or interesting findings you gathered from the literature. Share your notes with one team members.





Let's wrap up this session!

Questions? Final thoughts?

EXTRA SLIDES FOR DAY 4

Case Study: Leila's Smartphone Research

Research Question

"How does nighttime smartphone use relate to students' sleep quality and academic performance?"

Hypothesis

Students with more phone use after 10 p.m. will report lower sleep quality and worse academic performance.

Design Elements

- Correlational study using surveys
- 120 students from grades 10-12
- Sleep journals from 25 volunteers
- Follow-up interviews with 10 participants



Leila's study examined a real-world issue affecting many students.

Leila's Research Timeline

Weeks 1-2: Baseline Data

Distribute digital survey during advisory periods. Collect information on smartphone usage, sleep quality, and academic focus.



Weeks 3-4: Follow-up

Conduct 10 individual interviews with participants showing varying screen time behaviors.

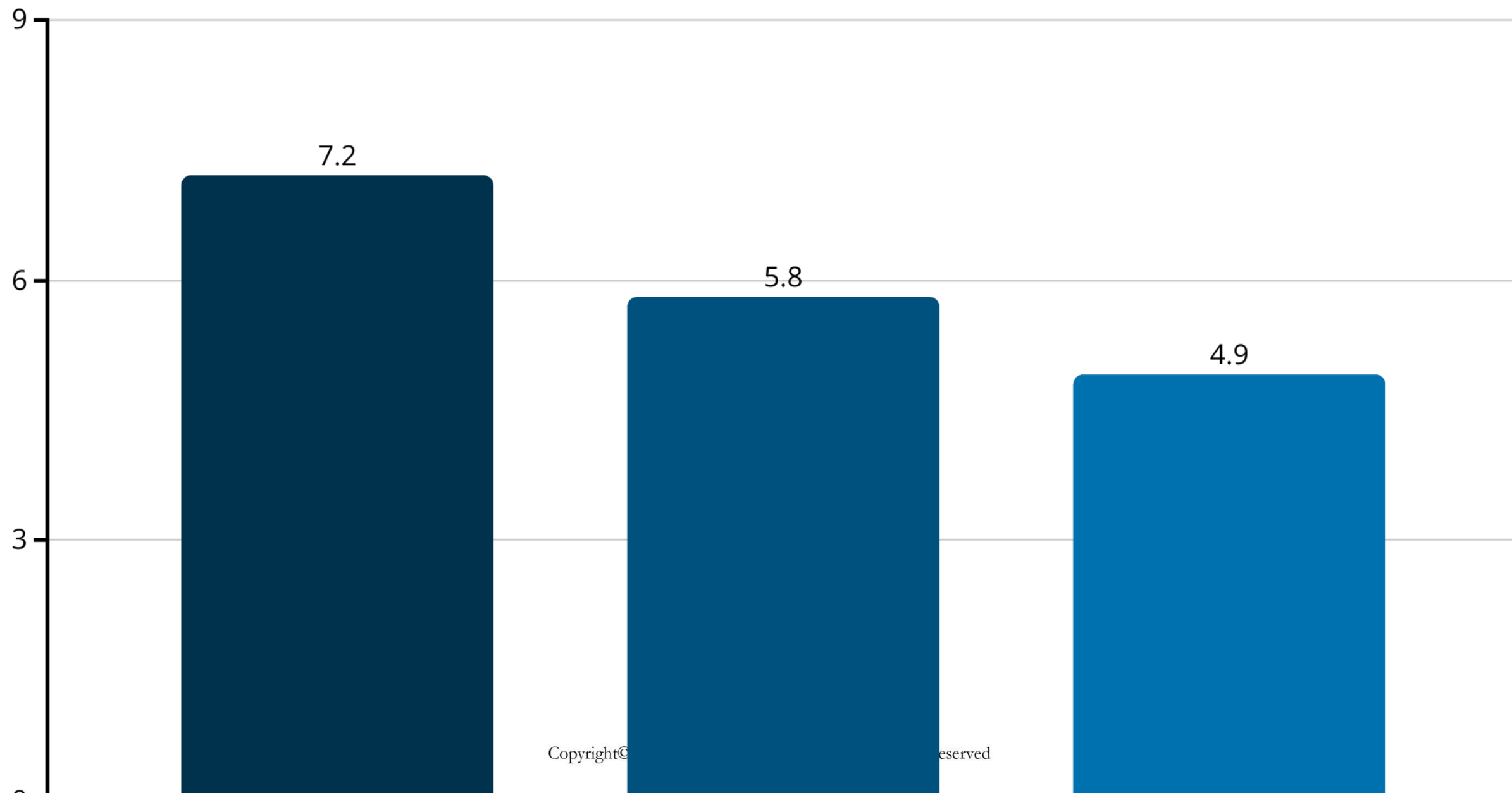


Week 5: Analysis

Code data, analyze results, and prepare presentation materials.



Leila's Findings





Leila's Research Impact

School Presentation

Shared findings at a health and wellness night, educating the community about digital habits.

Student Campaign

Inspired a student-led initiative promoting "screen-free wind-down hours" before bedtime.

Broader Conversation

Helped open discussion about digital wellness and healthy technology use among teens.

Next: Let's Experiment!



Hands-On Mini Experiment

Chewing Gum & Memory

Research Question: “Does chewing gum improve short-term memory in middle and high school students?”

Hypothesis (Testable Statement): “Students who chew gum will remember more words from a list than students who don’t.”

So let’s think about what kind of data we can work with? Qualitative vs Quantitative?

And how we can go about collecting our data to test our very testable statement aka our hypothesis?

Hands-On Mini Experiment

Chewing Gum & Memory

Research Question: “Does chewing gum improve short-term memory in middle and high school students?”

Hypothesis (Testable Statement): “Students who chew gum will remember more words from a list than students who don’t.”

Experiment Steps:

Here is a list of 10 simple words.

We are to randomly divide you all into two groups (chewing gum vs. no gum).

You will see the list for 30 seconds. Then we will wait for 1 minute. Then you will write as many words as you can remember.

That would allow us to compare average number of words recalled per group.

Hands-On Mini Experiment

Chewing Gum & Memory - Research Design Discussion:

Independent variable:

Gum or no gum

Dependent variable:

Number of words remembered

Control variables:

Word list, exposure time, wait time before recall, setting

Could the results support or refute the hypothesis?

What might influence that outcome?

e.g., nerves, distractions, small sample size

